

Cambridge (CIE) IGCSE Maths
Paper 1: Non-Calculator (Set E)
Core

Saturday 26 April 2025

Morning (Time: 1 hour 30 minutes)

Total marks

/ 83

Instructions

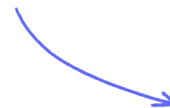
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You will need geometrical instruments.
- Calculators must not be used in this paper.

Materials

- [List of formulas](#)

Scan here to mark your mock exam

or visit the mock exams landing page for this course



1 A cruise ship travels 2067 km.

i) Write 2067 in words.

[1]

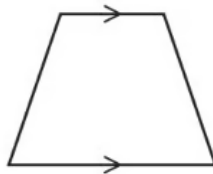
ii) Write 2067 correct to the nearest hundred.

[1]

(2 marks)

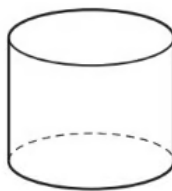
2 Write down the mathematical name for this

i) quadrilateral,



[1]

ii) solid.



[1]

(2 marks)

3 (a) Using the integers from 60 to 75 only, find

i) a multiple of 17,

[1]

ii) the prime numbers.

[2]

(3 marks)

(b) Write down the reciprocal of 7.

(1 mark)

4 Marco has ropes of four different colours. He takes a rope at random.

Colour	Brown	White	Red	Green
Probability	0.35		0.04	0.2

Complete the table.

(2 marks)

- 5 A box contains 22 coloured pencils.
6 pencils are pink, 9 pencils are blue and 7 pencils are yellow.

Write down the ratio pink pencils : not pink pencils.
Give your answer in its simplest form.

(2 marks)

- 6 The scale drawing shows the positions of town S and town T .
The scale is 1 centimetre represents 15 kilometres.



Scale: 1 cm to 15 km

- i) Find the actual distance between these two towns.

..... km [2]

- ii) Measure the bearing of town T from town S .

[1]

(3 marks)

7 Saskia has \$600.

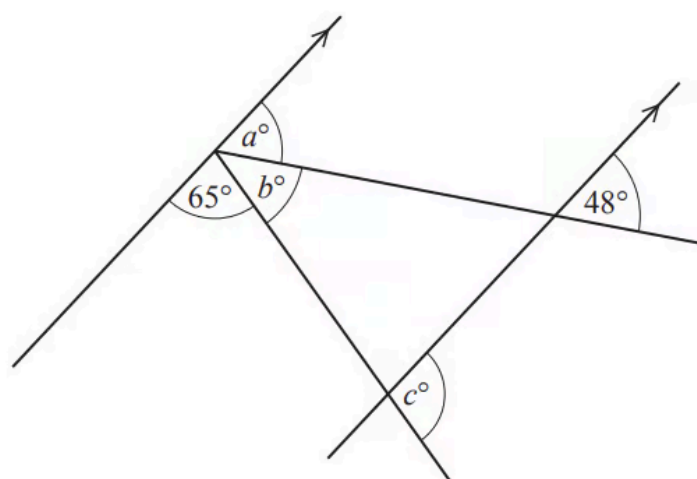
She spends $\frac{1}{5}$ of the \$600 on a coat and gives $\frac{1}{3}$ of the \$600 to her son.

What fraction of the \$600 does she have left?

Give your answer in its simplest form.

(3 marks)

8



NOT TO
SCALE

The diagram shows two parallel lines and two straight lines.

i) Find the value of a .

Give a reason for your answer.

$a = \dots\dots\dots$ because $\dots\dots\dots$ [2]

ii) Find the value of b .

Give a reason for your answer.

$b = \dots\dots\dots$ because $\dots\dots\dots$ [2]

iii) Find the value of c .

$$c = \dots\dots\dots [2]$$

(6 marks)

9 Without using a calculator, work out $2\frac{1}{3} + \frac{4}{5}$.

You must show all your working and give your answer as a mixed number in its simplest form.

(3 marks)

10 Put one pair of brackets into the calculation to make it correct.

$$8 + 6 - 2 \times 5 = 28$$

(1 mark)

11 Factorise $3x + 12$.

(1 mark)

12 Solve.

$$5(2x + 4) = 85$$

$x =$

(2 marks)

13 Find the number of minutes from 1758 to 7.13 pm.

..... min

(2 marks)

14 The length, l , of a rectangle is 3 cm longer than the width, w .

The perimeter of the rectangle is 26 cm.

Calculate the length, l , and the width, w .

$l =$ cm

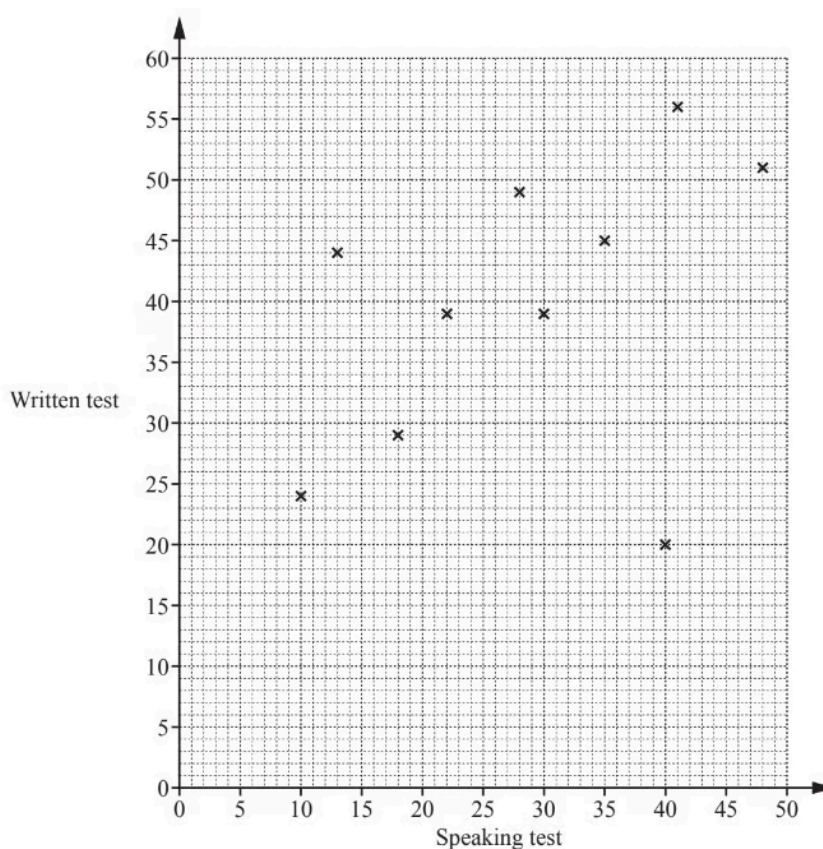
$w =$ cm

(3 marks)

- 15 (a)** Fourteen students each take two tests in French, a speaking test and a written test. The table shows the scores.

Speaking test	10	13	48	30	35	18	41	40	22	28	20	44	37	46
Written test	24	44	51	39	45	29	56	20	39	49	33	52	44	52

Complete the scatter diagram. The first ten points have been plotted for you.



(2 marks)

- (b)** What type of correlation is shown in this scatter diagram?

(1 mark)

- (c)** One student has a high score in the speaking test and a low score in the written test.

On the scatter diagram, put a ring around this point.

(1 mark)

- (d)** On the scatter diagram, draw a line of best fit.

(1 mark)

- (e)** Use your line of best fit to estimate a score in the written test for a student who scored 25 in the speaking test.

(1 mark)

16 (a)

27	14	8	93	32	55	14	38	73	47
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From this list of numbers, find the median.

[2]

(2 marks)

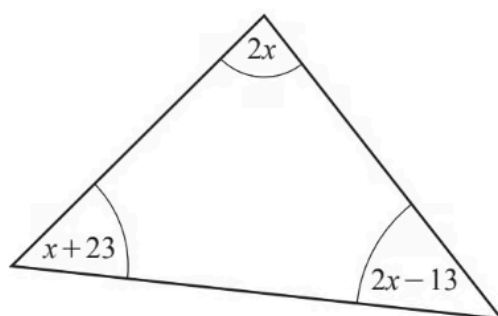
(b)

27	14	8	93	32	55	14	38	73	47
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From this list of numbers, find the range.

(1 mark)

17 In this part, all angles are in degrees.



NOT TO
SCALE

i) Use the information in the triangle to write down an equation in terms of x .

[1]

ii) Solve this equation to find the value of x .

$x = \dots\dots\dots$ [3]

iii) Work out the size of the smallest angle in the triangle.

[2]

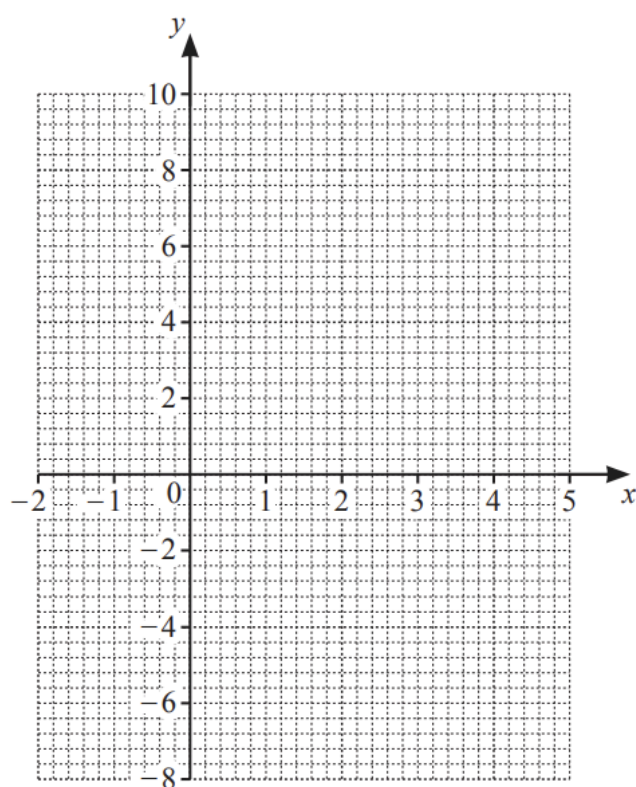
(6 marks)

18 (a) Complete the table of values for $y = x^2 - 4x - 3$

x	-2	-1	0	1	2	3	4	5
y		2	-3	-6		-6	-3	2

(2 marks)

(b) On the grid, draw the graph of $y = x^2 - 4x - 3$ for $-2 \leq x \leq 5$.



(4 marks)

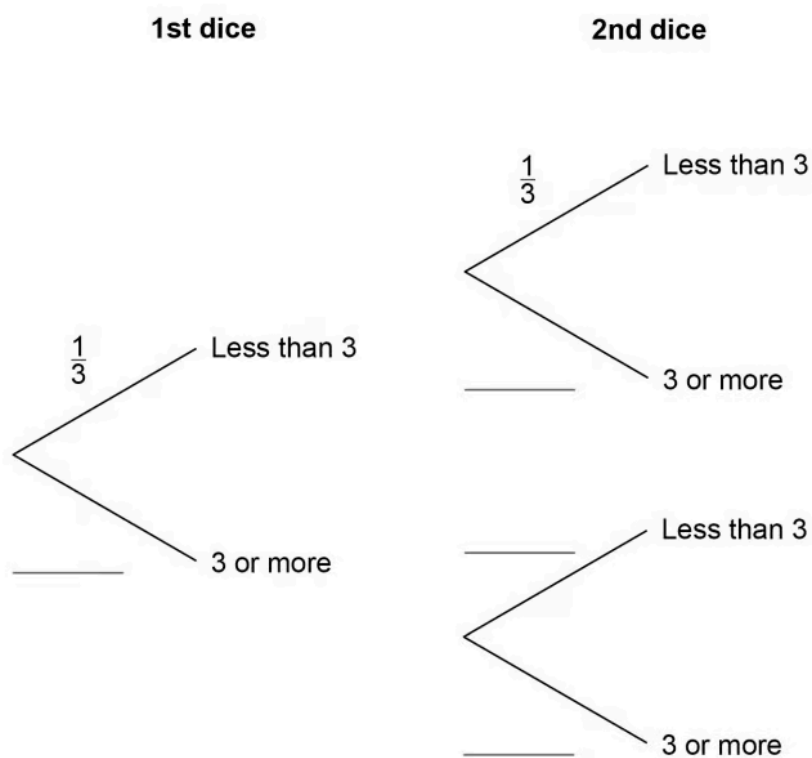
(c) Use your graph to solve the equation $x^2 - 4x - 3 = 0$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(2 marks)

19 (a) Two ordinary fair dice are rolled.

Complete the tree diagram.



(1 mark)

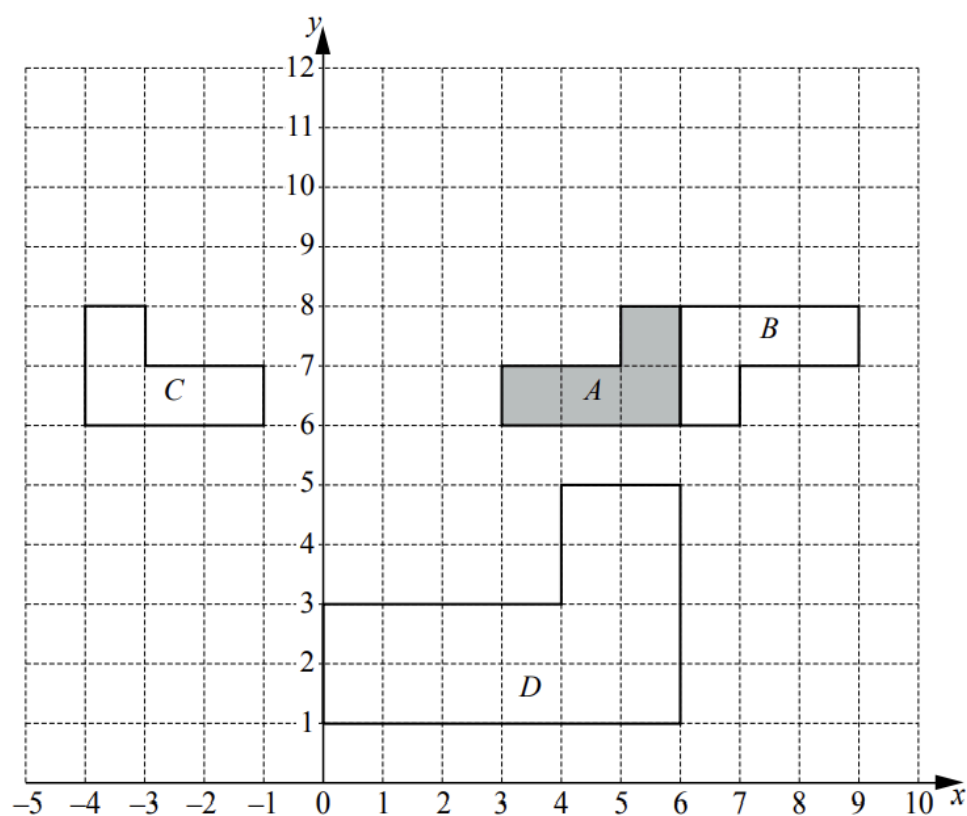
(b) Work out the probability that **both** dice land on a number less than 3.

(1 mark)

(c) Work out the probability that **exactly one** of the dice lands on a number less than 3.

(2 marks)

20 (a)



The diagram shows four shapes A , B , C and D .

Describe fully the **single** transformation that maps shape A onto

i) shape B ,

[3]

ii) shape C ,

[2]

iii) shape D .

[3]

(8 marks)

(b) On the grid, draw the image of shape A after a translation by the vector $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$

(2 marks)

21 Make m the subject of the formula $y = 4m - p$

$m = \dots\dots\dots$

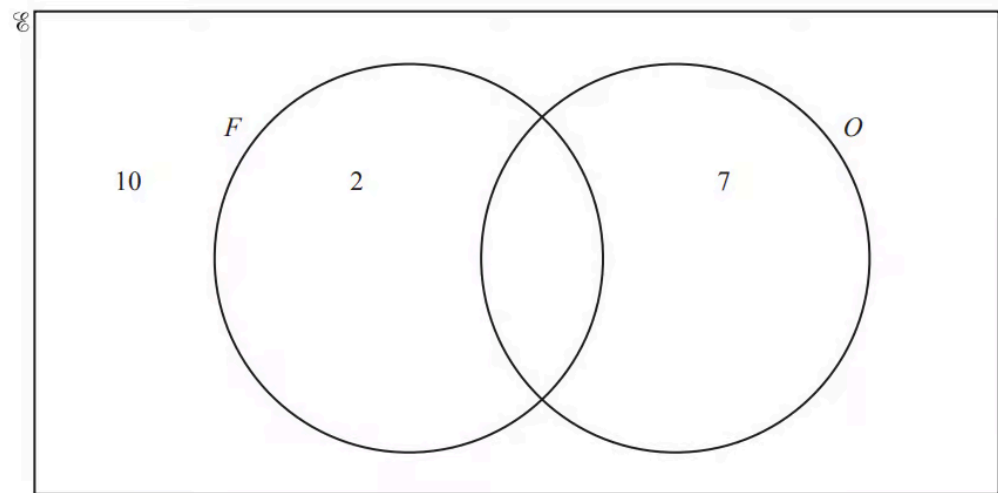
(2 marks)

22 $\mathcal{E} = \{x : x \text{ is a natural number } \leq 15\}$

$F = \{x : x \text{ is a factor of } 12\}$

$O = \{x : x \text{ is an odd number}\}$

i) Complete the Venn diagram to show the elements of these sets.



[2]

ii) Write down one number that is in set O , but not in set F .

[1]

iii) Find $n(F \cup O)$.

[1]

iv) A number is chosen at random from E .

Work out the probability that this number is in set O .

[1]

(5 marks)

23 The length of one side of a rectangle is 12cm.

The length of the diagonal of the rectangle is 13cm.

Calculate the area of the rectangle.

..... cm²

(3 marks)